

Multiple Choice (10 marks)

1. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
 - (a) Round-robin
 - (b) Priority queuing
 - (c) Shortest job first
 - (d) Youngest job first
2. Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is NOT complete?
 - (a) {AND, NOT}
 - (b) {NOT, OR}
 - (c) {AND, OR}
 - (d) {NAND}
3. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
4. Local caching of files is common in distributed file systems, but it has the disadvantage that
 - (a) Temporary inconsistencies among views of a file by different machines can result
 - (b) The file system is likely to be corrupted when a computer crashes
 - (c) A much higher amount of network traffic results
 - (d) Caching makes file migration impossible
5. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: The regular expression $a^*(c + d)^* + b(c + d)^*$ is equivalent to $(a^* + b)^*(c + d)$ and describes the same set of strings.
7. True or False: The value of the Python expression $4 + 3 \% 2$ is 5, as the modulus operator is evaluated before addition.
8. True or False: The output of the statement `'a' + 'ab'` in Python 3 is `'aab'`.
9. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is n^m .
10. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][1]` evaluates to the integer 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the two numbers differ at one bit position only or not.
12. Write a function to convert more than one list to nested dictionary.

End of Paper